



06B THERMOCHEMISTRY

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 6

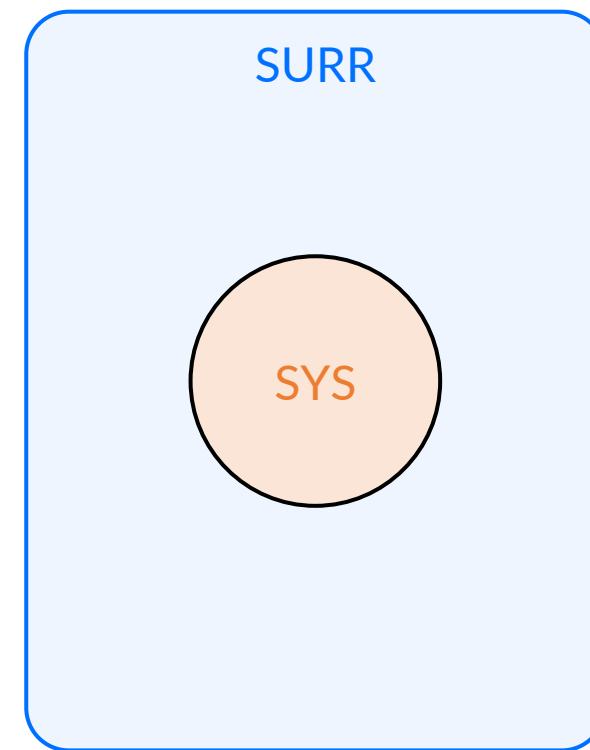
1. Define and apply energy, work, and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state functions and path functions.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Summarize the roles of enthalpy and heat flow in chemical reactions.
6. Calculate pressure-volume work.
7. Calculate the heat required to change the temperature of a substance.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of different reactions.
10. Apply standard enthalpies of formation and Hess's law to calculate the enthalpy of a reaction.

Energy and Enthalpy

6.4

The change in enthalpy for a reaction (ΔH_{rxn}) is *defined* as the flow of heat (q) when the chemical reaction is carried at a constant (atmospheric) pressure.

Process	System/Rxn	Surroundings	Sign of ΔH_{rxn}
Exothermic	<i>Releases heat</i>	<i>Absorbs heat</i>	ΔH_{rxn}
Endothermic	<i>Absorbs heat</i>	<i>Releases heat</i>	ΔH_{rxn}



Energy and Enthalpy

Chemistry in Context

Recall that heat (q) is the transfer of energy *due to a temperature difference* between two objects. Heat will always flow until the two objects reach the same temperature, or a state called thermal equilibrium.

You should recognize that heat is an incredibly easy quantity for us to measure in the lab because it is related to changes in temperature. All we need is a thermometer!

The next several sections are all related to *different ways* to measure or calculate ΔH_{rxn} .

There are three general methods:

- 1) Experimentally through calorimetry (measuring temperature changes)
- 2) Hess's Law: taking advantage of state function properties
- 3) Theoretically through standard enthalpies of formations, ΔH_f°

Specific Heat

6.5

In the *absence* of a phase change, the amount of heat (q) required to change the temperature (ΔT) of some quantity (m) of substance is given by

$$q = m \cdot c \cdot \Delta T$$

Specific heat capacity: (also called *specific heat*, c)

- The amount of energy required to increase the temperature of *exactly* 1 g of a substance by *exactly* 1 °C = 1 K.
- Also comes in a per mole variety called the *molar heat capacity*.

Specific Heat

6.5

- Notes:
- 1) Liquid water has a *large* value of c .
 - 2) Many metals tend to have *small* values of c .
 - 3) We assume c does not change with temperature (not technically true).

Covalent Substance	Specific Heat $\left(\frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}\right)$
CO ₂ (g)	0.852
CO (g)	1.04
H ₂ (g)	14.4
N ₂ (g)	1.04
O ₂ (g)	0.922
H ₂ O (g)	2.042
H ₂ O (ℓ)	4.184
H ₂ O (s)	2.089

Metal Substance	Specific Heat $\left(\frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}\right)$
Au (s)	0.129
Pb (s)	0.13
Sn (s)	0.22
Ag (s)	0.24
Na (s)	0.293
Cu (s)	0.385
Zn (s)	0.388
Fe (s)	0.442
Cr (s)	0.45
Co (s)	0.46
Al (s)	0.892
Mg (s)	1.0

TABLE 6.3 Specific Heat Capacities of Selected Substances

EXAMPLE PROBLEM

Learning Objective(s): 6.7

What is the specific heat capacity (c) of an unknown metal if it takes 1155 J of heat to raise the temperature of 100.0 g of the metal from 25.0 °C to 55.0 °C?

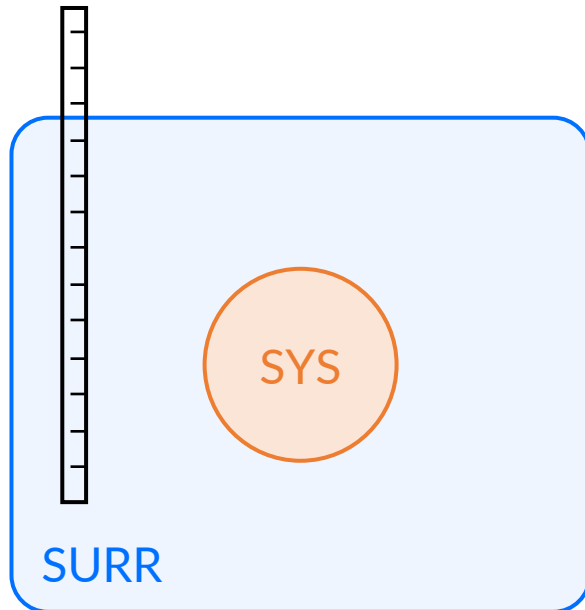
What is the identity of the metal?

Calorimetry: Measuring Energy Changes

6.6

Calorimetry: The study of heat transfer between substances by measuring the temperature changes of the substances involved.

- The heat released or absorbed by the **system** is determined by measuring the temperature change of the **surroundings**.



IN-CLASS QUESTION 1

Learning Objective(s): 6.7

A 25.0 g block of metal initially at 95.0 °C is placed in a calorimeter containing 121 g of water initially at 27.2 °C. Assuming that no heat is lost to the calorimeter or the surroundings, what is the final temperature (in °C) of the water and metal?

$$c_{\text{metal}} = 0.49 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

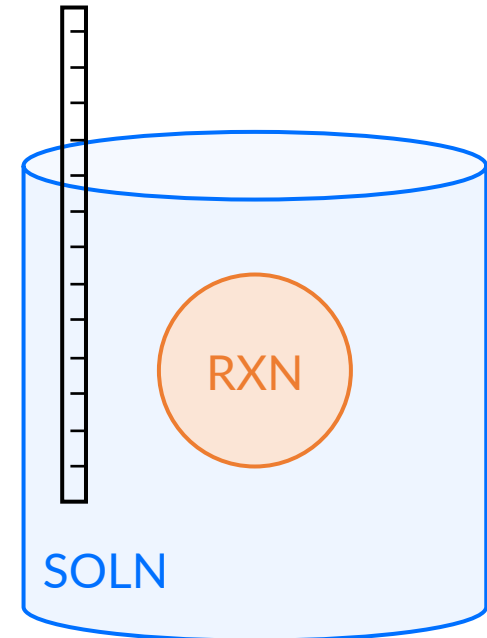
$$c_{\text{water}} = 4.184 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

Calorimetry: Measuring Energy Changes

6.6

Constant pressure calorimetry takes place on the benchtop where the atmospheric pressure is relatively constant such that $q_{P,\text{rxn}}$ (and ΔH_{rxn}) can be measured.

- The **reaction (rxn)** is the **system** and the **solution (soln)** in which the reaction takes place is the **surroundings**.

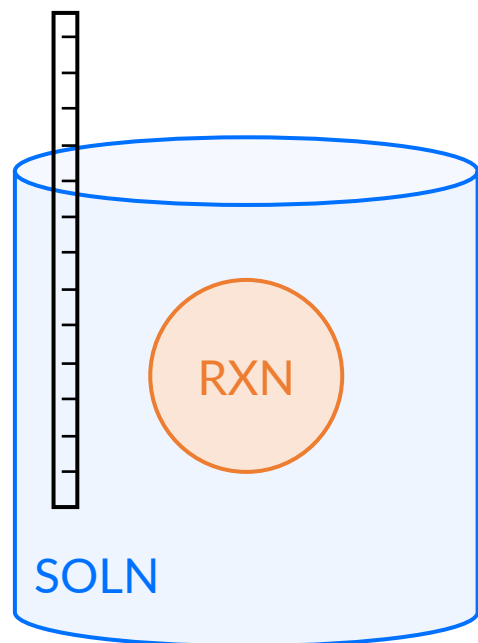


Calorimetry: Measuring Energy Changes

6.6

Remember that “system” and “surroundings” are fictitious designations.

In an *aqueous* reaction, we measure the temperature of the solution and the solution is part of the surroundings.



Temperature of solution	Surroundings (soln)	System (rxn)	ΔH_{rxn}
Increases	absorbs heat	releases heat	ΔH_{rxn}
Decreases	releases heat	absorbs heat	ΔH_{rxn}

IN-CLASS QUESTION 2

Learning Objective(s): 6.7

A 30.0 mL sample of 0.500 M NaOH and a 10.0 mL sample of 0.500 M HCl are combined in a constant pressure calorimeter. The temperature of the combined solutions increases from 20.40 °C to 23.55 °C. Assuming the calorimeter loses no heat to the surroundings, what is ΔH_{rxn} in kJ/mol?

Assume the density of the solution is 1 g/mL and the specific heat capacity is that of water.

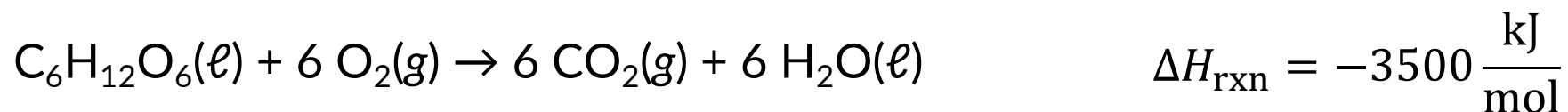
Enthalpy in Chemical Reactions

6.7

Thermochemical equation

- Coupling a balanced chemical equation with its corresponding *change in enthalpy*, ΔH_{rxn} (also called *heat of reaction* or *enthalpy of reaction*).
- Thermochemical equations are *only* interpreted as mole-mole ratios.
- For example, we'd say that:

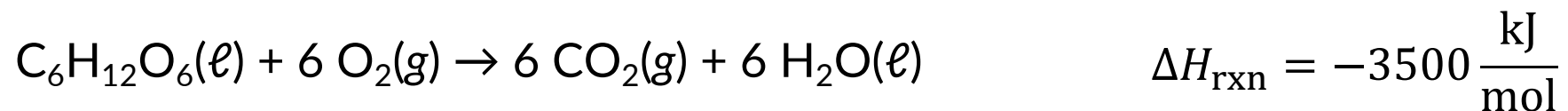
“For every one mole of reaction, one mole of $\text{C}_6\text{H}_{12}\text{O}_6$ reacts with six moles of O_2 to produce six moles of CO_2 and six moles of H_2O , and releases 3500 kJ of heat.”



Enthalpy in Chemical Reactions

6.7

Thermochemical equations are also *only* interpreted to represent the enthalpy change for the reaction ***exactly as it is written from left to right.***



- The units for ΔH_{rxn} are kJ/mol or J/mol and interpreted as the quantity of heat per mole of reaction (rxn).
- ΔH_{rxn} is an *intensive property* specific to a reaction, so it is a constant.
- If you carry out more or less than one mole of reaction, then the quantity of heat (q_{rxn}) associated with the experiment carried out changes/scales.

IN-CLASS QUESTION 3

Learning Objective(s): 6.8

The overall reaction in a commercial heat pack can be represented as:

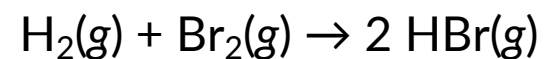


What is the total heat (q_{rxn} in kJ) associated with the reaction between 2.00 g of Fe(s) and *excess* O₂(g)?

EXAMPLE PROBLEM

Learning Objective(s): 6.8

How much energy, as heat in kJ, would be needed to *decompose* 9.74 g of HBr(g) into its elements?



$$\Delta H_{\text{rxn}}^{\circ} = -72.8 \frac{\text{kJ}}{\text{mol}}$$

BONUS QUESTION

Assessed on: *undetermined*

The overall reaction in a commercial heat pack can be represented as:



What is the total heat (q_{rxn} in kJ) associated with the reaction between 515 g of Fe(s) and 475 g O₂(g)?